16. $s = \sqrt{MSE} = \sqrt{41.25} = 6.422$

$SE = \frac{s}{\sqrt{n_i}} = \frac{6.422}{\sqrt{5}} = 2.872$

$q_{0.01}(4,16) \approx 5.19$

$w = q \cdot SE = 14.906$

$|\bar{x}_3 - \bar{x}_2| = |251.6 - 234.8| = 16.8$

$|\bar{x}_1 - \bar{x}_3| = |262.2 - 251.6| = 10.6$

$|\bar{x}_4 - \bar{x}_1| = |268.6 - 262.2| = 6.4$

State 2 is significantly different from all other.

State 3 is significantly difference from the mean cost in state 4.

17. (a) $H_0: \mu_1 = \mu_2 = \mu_3$

H_a: 至少一个 mean is different

Source	df	SS	MS	F
Treatment	2	34806.67	37403.33	4.47
Error	27	225930	8367.78	
Total	29	300736.67		

$4.47 > F_{0.05,2,27} \approx 3.35$

$\Rightarrow$ reject H_0

(b) 95% CI for $\mu_1 - \mu_2$

$\Sigma x_1 = 6080 \quad \Sigma x_2 = 6530 \quad MSE = 8367.78$

$s \approx \sqrt{MSE} = 91.4756$

$\bar{x}_1 = \frac{\Sigma x_1}{n_1} = \frac{6080}{10} = 608 \quad \bar{x}_2 = \frac{6530}{10} = 653$

$df = n - k = 27$

$t_{\alpha/2} = 1.96$

$E = t_{\alpha/2} \cdot s \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 1.96 \cdot 91.4756 \cdot \sqrt{\frac{2}{10}} = 80.1819$

$(\bar{x}_1 - \bar{x}_2) \pm E = (608 - 653) \pm 80.1819 = (-125.1819, 35.1819)$

(c) $k = 3 \quad n_t = 10 \quad df = 27 \quad q_\alpha(k, df) = q_{0.05}(3, 24) = 3.53$

$w = 3.53 \cdot \frac{91.4756}{\sqrt{10}} \approx 102.1127$

$\bar{x}_3 = 532$

$\bar{x}_3 < \bar{x}_1 < \bar{x}_2$

$|\bar{x}_3 - \bar{x}_1| = 76 < w$

$|\bar{x}_1 - \bar{x}_2| = 45 < w$

$\rightarrow$ Not significant

$|\bar{x}_2 - \bar{x}_3| = 121 > w \rightarrow$ significant

Significantly difference

24

(a) Treatment: Store Blocks: item

(b) $p = 0.100 < 0.05$ reject H_0

(c)

25.

(a) $k = 4$ (groups) $b = 8$ (blocks)

$df_E = 21$ $\alpha = 0.05$ $q_\alpha(k, df) = 3.96$

(b) $MSE = 0.1658$

$s = \sqrt{MSE} = 0.4071$

$w = q_\alpha(k, df) \frac{s}{\sqrt{b}} = 0.57$

26.